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In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create a sample diabetes dataset
data = {
    'Pregnancies': [6, 1, 8, 1, 0, 5, 3, 10, 2, 4],
    'Glucose': [148, 85, 183, 89, 137, 116, 78, 115, 197, 110],
    'BloodPressure': [72, 66, 64, 66, 40, 74, 50, 0, 70, 92],
    'SkinThickness': [35, 29, 0, 23, 35, 0, 32, 0, 45, 0],
    'Insulin': [0, 0, 0, 94, 168, 0, 88, 0, 543, 0],
    'BMI': [33.6, 26.6, 23.3, 28.1, 43.1, 25.6, 31.0, 35.3, 30.5, 37.6],
    'DiabetesPedigreeFunction': [0.627, 0.351, 0.672, 0.167, 2.288, 0.201, 0.248, 0
    'Age': [50, 31, 32, 21, 33, 30, 26, 29, 53, 30],
    'Outcome': [1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
}

diabetes_df = pd.DataFrame(data)

# 1. Display the first five rows
print("First Five Rows:")
print(diabetes_df.head())

# 2. Display the last five rows
print("\nLast Five Rows:")
print(diabetes_df.tail())

# 3. Display dataset information
print("\nDataset Information:")
diabetes_df.info()

# 4. Display the number of rows and columns
print("\nDataset Shape:")
print(diabetes_df.shape)

# 5. Display column names
print("\nColumn Names:")
print(diabetes_df.columns)

# 6. Check for missing values
print("\nMissing Values:")
print(diabetes_df.isnull().sum())

# 7. Display statistical summary
print("\nStatistical Summary:")
print(diabetes_df.describe())

# 8. Check duplicate rows
print("\nNumber of Duplicate Rows:")
print(diabetes_df.duplicated().sum())

# 9. Visualize the correlation between columns
plt.figure(figsize=(10, 6))
```

```
sns.heatmap(diabetes_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title("Correlation Heatmap of Diabetes Dataset")
plt.show()

# 10. Visualize the outcome distribution
sns.countplot(data=diabetes_df, x='Outcome')
plt.title("Diabetes Outcome Distribution")
plt.xlabel("Outcome (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("Count")
plt.show()
```

```
First Five Rows:
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1

Last Five Rows:
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
5            5      116             74              0        0  25.6
6            3       78             50             32       88  31.0
7           10      115              0              0        0  35.3
8            2      197             70             45      543  30.5
9            4      110             92              0        0  37.6

   DiabetesPedigreeFunction  Age  Outcome
5                     0.201   30        0
6                     0.248   26        1
7                     0.134   29        0
8                     0.158   53        1
9                     0.191   30        0

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               10 non-null     int64
 1   Glucose                   10 non-null     int64
 2   BloodPressure             10 non-null     int64
 3   SkinThickness             10 non-null     int64
 4   Insulin                   10 non-null     int64
 5   BMI                       10 non-null     float64
 6   DiabetesPedigreeFunction  10 non-null     float64
 7   Age                       10 non-null     int64
 8   Outcome                   10 non-null     int64
dtypes: float64(2), int64(7)
memory usage: 852.0 bytes

Dataset Shape:
(10, 9)

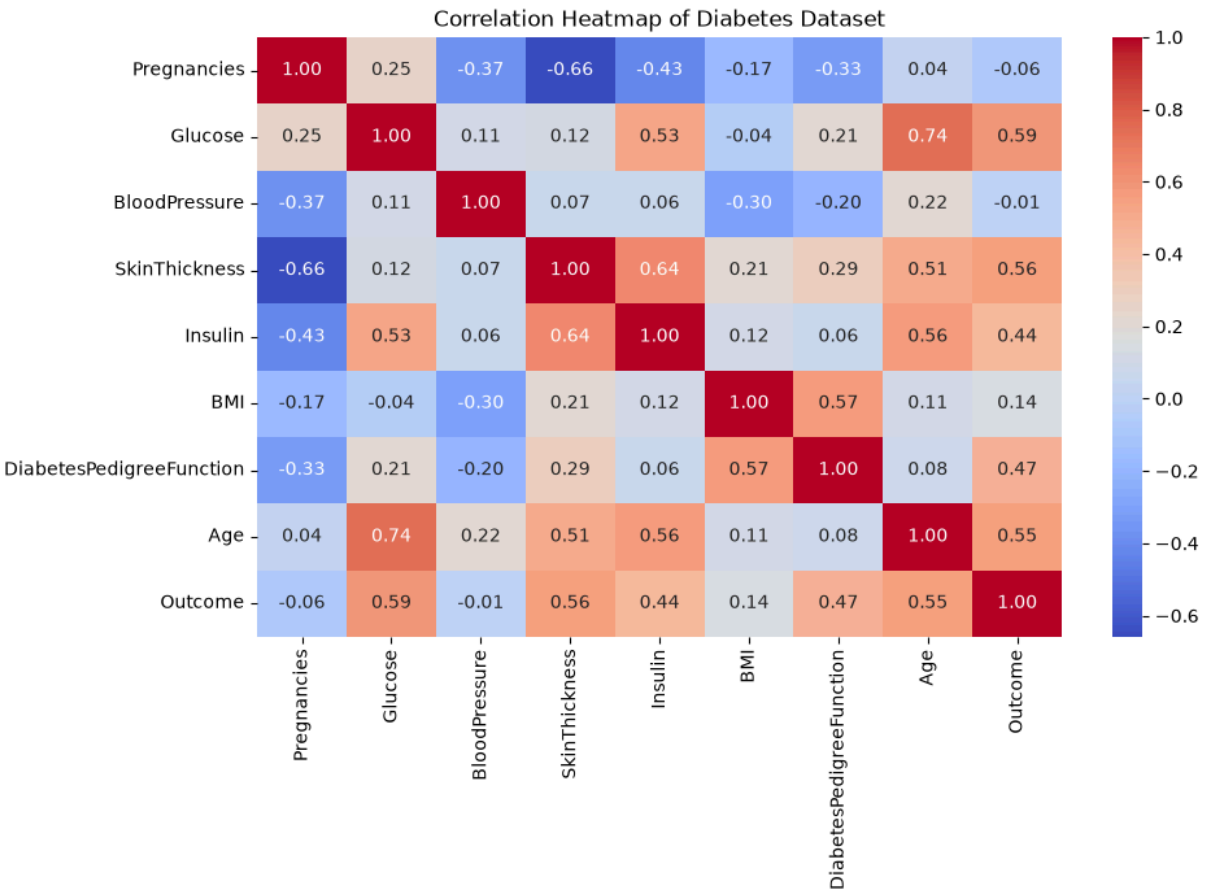
Column Names:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='str')
```

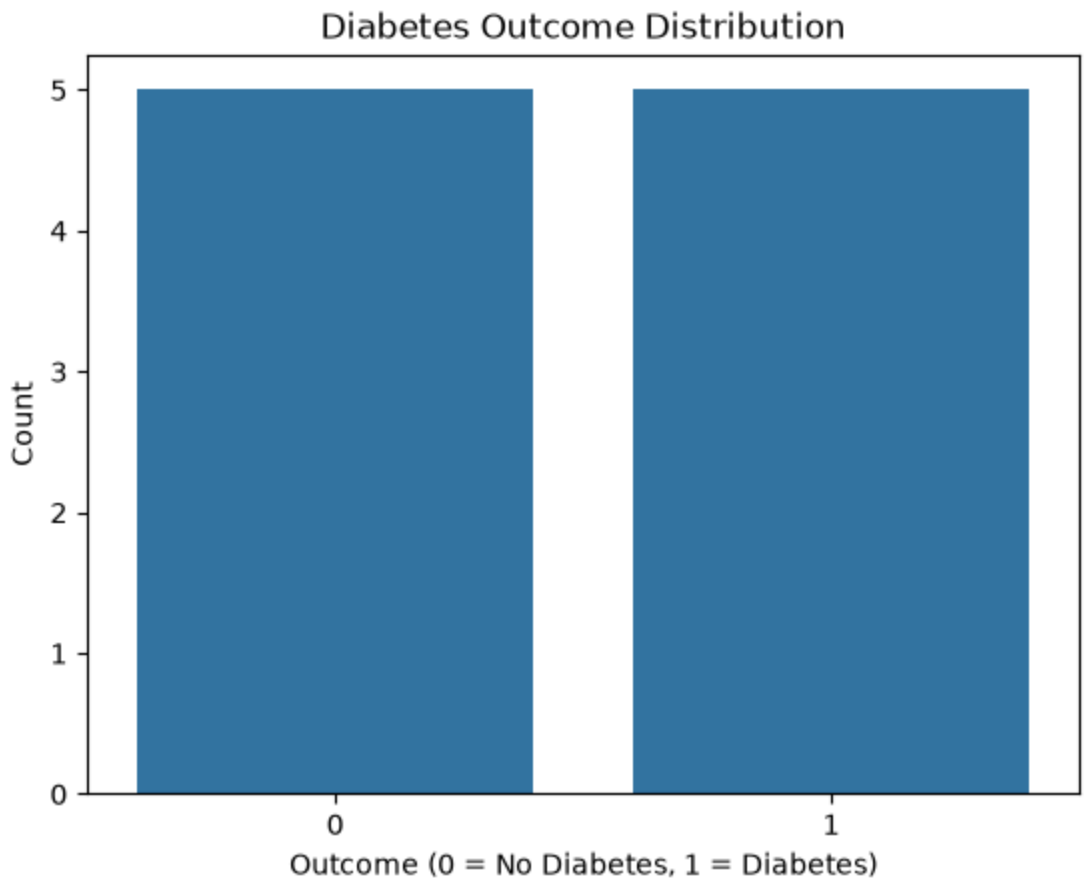
```
Missing Values:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Statistical Summary:
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count    10.000000   10.000000      10.000000      10.000000   10.000000
mean      4.000000  125.800000      59.400000      19.900000   89.300000
std       3.265986   40.449694      25.051946      17.978073  169.937276
min       0.000000   78.000000       0.000000       0.000000    0.000000
25%       1.250000   94.250000      53.500000       0.000000    0.000000
50%       3.500000  115.500000      66.000000      26.000000    0.000000
75%       5.750000  145.250000      71.500000      34.250000   92.500000
max      10.000000  197.000000      92.000000      45.000000  543.000000

             BMI  DiabetesPedigreeFunction        Age    Outcome
count  10.000000                  10.00000  10.000000  10.000000
mean   31.470000                   0.50370  33.500000   0.500000
std     6.050537                   0.65616  10.102255   0.527046
min    23.300000                   0.13400  21.000000   0.000000
25%    26.975000                   0.17300  29.250000   0.000000
50%    30.750000                   0.22450  30.500000   0.500000
75%    34.875000                   0.55800  32.750000   1.000000
max    43.100000                   2.28800  53.000000   1.000000

Number of Duplicate Rows:
0
```


Correlation Heatmap of Diabetes Dataset
Pregnancies 1.00 0.25 -0.37 -0.66 -0.43 -0.17 -0.33 0.04 -0.06
Glucose 0.25 1.00 0.11 0.12 0.53 -0.04 0.21 0.74 0.59
BloodPressure -0.37 0.11 1.00 0.07 0.06 -0.30 -0.20 0.22 -0.01
SkinThickness -0.66 0.12 0.07 1.00 0.64 0.21 0.29 0.51 0.56
Insulin -0.43 0.53 0.06 0.64 1.00 0.12 0.06 0.56 0.44
BMI -0.17 -0.04 -0.30 0.21 0.12 1.00 0.57 0.11 0.14
DiabetesPedigreeFunction -0.33 0.21 -0.20 0.29 0.06 0.57 1.00 0.08 0.47
Age 0.04 0.74 0.22 0.51 0.56 0.11 0.08 1.00 0.55
Outcome -0.06 0.59 -0.01 0.56 0.44 0.14 0.47 0.55 1.00
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6


Diabetes Outcome Distribution
Count
0
1
2
3
4
5
0
1
Outcome (0 = No Diabetes, 1 = Diabetes)

In [ ]: